

AI in E-Commerce: Enhancing Customer Experience

Slide 1: Title Slide

- ▶ Topic: AI in E-Commerce
- ▶ Focus: Enhancing Customer Experience
- ▶ Story: Anita (shopper) meets AI (smart assistant)
- ▶ Objective: Understand AI from a research perspective
- ▶ **Hook:** E-commerce without AI is like Netflix without recommendations.

Slide 2: Problem Statement

- ▶ Customers face:
 - ▶ Too many choices (decision fatigue)
 - ▶ Irrelevant recommendations
 - ▶ Poor support experience
- ▶ Businesses face low retention and conversions
- ▶ **Story:** Anita searches for shoes but gets random results
- ▶ **Analogy:** Asking for pizza and getting salad instead
- ▶ **Research Gap:** Lack of real-time personalization

Slide 3: Methodology

- ▶ Data collection (clicks, behavior, history)
- ▶ Data preprocessing (cleaning and structuring)
- ▶ Model training (AI/ML algorithms)
- ▶ Deployment in platforms
- ▶ Continuous feedback loop
- ▶ **Story:** AI observes Anita's behavior silently
- ▶ **Analogy:** A shopkeeper who remembers everything

Slide 4: Data Description

- ▶ Types of data:
 - ▶ Purchase history
 - ▶ Browsing behavior
 - ▶ Search queries
 - ▶ Reviews and ratings
- ▶ Structured and unstructured data
- ▶ **Story:** AI builds a digital personality of Anita
- ▶ **Analogy:** Like a best friend who knows your taste
- ▶ **Research Insight:** Better data leads to better predictions

Slide 5: Model / Approach

- ▶ Recommendation systems (collaborative, content-based)
- ▶ NLP-based chatbots
- ▶ Predictive analytics
- ▶ Dynamic pricing models
- ▶ **Story:** AI suggests perfect products instantly
- ▶ **Analogy:** Mind-reading salesperson
- ▶ **Research Contribution:** Hybrid models improve accuracy

Slide 6: Results

- ▶ Increased customer satisfaction
- ▶ Higher conversion rates
- ▶ Reduced cart abandonment
- ▶ Faster product discovery
- ▶ **Story:** Anita finds products quickly and buys more
- ▶ **Analogy:** From jungle search to GPS navigation
- ▶ **Research Finding:** AI improves efficiency and engagement

Slide 7: Conclusion

- ▶ AI enables:
 - ▶ Personalization
 - ▶ Efficiency
 - ▶ Predictive systems
- ▶ **Story:** Anita no longer searches, AI delivers
- ▶ **Analogy:** Maze turned into guided tour
- ▶ Future: Voice commerce, visual search, emotion-aware AI
- ▶ **Key Insight:** AI builds relationships, not just sales

Slide 8: Q&A

- ▶ Questions and discussions
- ▶ Topics:
 - ▶ Privacy vs personalization
 - ▶ Ethical concerns
 - ▶ AI limitations
- ▶ **Fun Line:** AI predicts shopping... can it predict your questions?